
Improved Decoding Algorithms for k-NAF Protection

K Kevin Sona
DA24B007

Kirthan S
DA24B009

Harshvardhan Agrawal
DA24B039

Abstract

With remarkable progress in LLMs also comes major privacy threats, because of the large corpora of web-crawled data used for pre-training, which invariably contain copyright-protected works. Due to the large proportions of such protected text in pre-training, LLMs invariably tend to emit large amounts of *verbatim* spans. We propose *Logit-Aggregate algorithm*, which is an inference time method that takes logits from a safe model trained on open datasets and a risky model trained on copyrighted dataset, and computes a new aggregate logit, from which we sample tokens. We also construct an objective function which is optimized by our choice of algorithm and prove strong privacy guarantees for the same. We then provide experimental results comparing our algorithm with Anchored Decoding.

1 Introduction

Copyright-safety in LLM inference has become increasingly important due to verbatim output of copyright data. Procuring licenses for data is infeasible due to the sheer scale of data involved. Filtering data into copyrighted and open data itself can be a difficult problem to tackle, since the definition and extent of copyrighted data can differ with context. Even if data is filtered, models trained purely on safe data perform very poorly. This rises the need to use risky data for training, but not output it.

The logit-aggregate algorithm aims to tackle this problem by interpolating the logits of a safe model and a risky model. The safe model is trained exclusively on public domain and openly licensed text, while the risky model is trained on mixed-license sources that may include copy righted or otherwise sensitive documents. The goal is to combine the desirable features from each model, privacy guarantees from the safe model and utility from the risky model.

Formally, we aggregate the logits of the safe model and the clipped difference of the safe and risky logits, to obtain a new logit, whose softmax gives us a distribution over next-tokens satisfying privacy guarantees as described by Vyas et al. [1]. Then we move on to construct an optimization problem to which the proposed algorithm is the optimal solution, which allows us to do more theoretical analysis on it.

To gauge the utility of this method, we compare its performance to that of anchored decoding algorithm (He et al. [2]) satisfying identical guarantees. For the experimental set-up, we choose TinyComma 1.8B for our safe model and Llama 3.1 - 8B as our risky model. Note that both models share the same tokenizer, allowing us to aggregate the logits with ease. Hyperparameters include α , β , and clip value C . Each of them was tuned and have also been provided with heuristics.

2 Preliminaries

Describing the setting of our problem as follows: Let ϕ_s and ϕ_r be the logits of the safe and risky model. In other words, $P_{\text{risky}} = \text{softmax}(\phi_r)$ and $P_{\text{safe}} = \text{softmax}(\phi_s)$. Here P is the next-token distribution over vocabulary V , given a prompt x

$$P(y_{0:T-1} | x) = \prod_{t=0}^{T-1} p(y_t | y_{<t}, x) \quad \text{for } T \leq T_{\max}$$

We borrow our copyright-safety framework from the K-NAF paper, which captures the similarity between 2 distributions, described as follows

K-Near Access-Freeness (K-NAF) . Bounds the total divergence over the entire distribution of sequences generated by a model P relative to a safe model P_s

Definition 2.1. We say that model P is safe relative to P_s if for all input sequence x and every $T \leq T_{\max}$,

$$\Delta(P(\cdot|x) \parallel P_s(\cdot|x)) \leq k_x \quad (1)$$

where Δ is a divergence measure of choice, K is the total privacy budget and $P(\cdot|x)$ is shorthand for $P(y_{<T}|x)$. Our choice of Δ is the KL Divergence (Δ_{KL})

Our method draws inspiration from 2 other key papers, namely anchored decoding (He et al. [2]) and exponential mechanism (Vinod et al. [3]), which have been briefly discussed below. Anchored decoding provides a strong baseline for us to build upon and invisibleink motivates us to operate on the logits directly instead of the distributions.

Anchored decoding. We assume access to a safe model p_s trained only on permissively licensed text, and a more performant risky model p_r that may reproduce copyrighted data due to its broader, unchecked training set. Our goal is to construct a new distribution p^* that approximates p_r to maximize utility, while remaining strictly subject to the safety constraint relative to p_s . This yields the constrained sequence-level optimization:

$$\begin{aligned} p^* &= \arg \min_p D_{KL}(p(\cdot | x) \parallel p_r(\cdot | x)) \\ \text{s.t. } &D_{KL}(p(\cdot | x) \parallel p_s(\cdot | x)) \leq K \end{aligned} \quad (2)$$

where K is the total privacy budget. The above optimization problem is not easy to solve since the distribution is over the entire vocabulary, making our search space size $V^{T_{\max}}$. Hence the global objective is decomposed into a series of per-token objective as follows,

$$\begin{aligned} p_t^*(\cdot | y_{<t}, x) &= \arg \min_{p \in \Delta(\mathcal{V})} D_{KL}(p \parallel p_r(\cdot | y_{<t}, x)) \\ \text{s.t. } &D_{KL}(p \parallel p_s(\cdot | y_{<t}, x)) \leq k_t, \\ &\sum_{y \in \mathcal{V}} p(y) = 1, \quad p(y) > 0 \quad \forall y \in \mathcal{V}. \end{aligned} \quad (3)$$

where k_t is the per token budget, the sum of which is K , the total budget. This problem admits a closed-form solution that can be efficiently computed at each decoding step:

For a given local budget k_t at decoding step t , the optimal distribution p_t^* that solves Eq. (2) is a weighted geometric mean:

$$p_t^* = \frac{1}{Z} p_s(\cdot | y_{<t}, x)^{\frac{\lambda}{1+\lambda}} p_r(\cdot | y_{<t}, x)^{\frac{1}{1+\lambda}} \quad (4)$$

where Z is a normalization constant and $\lambda \geq 0$ is the dual variable (Lagrange multiplier) associated with the KL constraint, solved for by the Newton-Raphson algorithm.

Exponential mechanism. The setting here is slightly different from ours. It generates the next token $x_t \in V$ via the canonical exponential mechanism over the next-token logits with a set of privacy-sensitive references $R = (r_1, \dots, r_B) \in (V^*)^B$. Given a query q and sensitive references R , obtain the next-token logits

$$\phi_i = \phi(\cdot \mid q, r_i, x_{<t}) \in \mathbb{R}^{|V|} \text{ for each } r_i \in R$$

Clip and aggregate the logits as:

$$\bar{\phi} = \frac{1}{B} \sum_{i=1}^B \text{clip}_C(\phi_i)$$

We then sample a token $x_t \in V$ from the distribution:

$$Q(x_t \mid q, R, x_{<t}) \propto \exp \left(\frac{\sqrt{2\rho}}{\text{sens}(\bar{\phi})} \cdot \bar{\phi}(x_t) \right) \quad (5)$$

where $\text{sens}(\bar{\phi})$ measures the maximum change in $\bar{\phi}(y)$ when moving to an adjacent set R' . This mechanism gives us ρ -zCDP guarantee. We will be borrowing this result in future proofs.

3 Logit-aggregate algorithm

Let ϕ_{risky} and ϕ_{safe} be the logits of LLMs trained on risky and safe datasets respectively i.e, $P_{\text{risky}} = \text{softmax}(\phi_{\text{risky}})$ and $P_{\text{safe}} = \text{softmax}(\phi_{\text{safe}})$. Define the aggregation of these logits to obtain a new logit $\phi_\star$ as follows:

$$\phi_\star = \alpha\phi_{\text{safe}} + \beta\text{clip}_C(\phi_{\text{risky}} - \phi_{\text{safe}}), \quad (6)$$

where α, β, C are hyperparameters and $\text{clip}_C(v)$ clips the entries of v at C . Next we move onto defining an objective function for this solution, an intuitive explanation for the choice of that objective and then prove privacy guarantees for this proposed method.

Algorithm 1 LOGIT-AGGREGATE($\Phi_r, \phi_s, K, T_{\max}$)

- 1: **Input:** risky Logit ϕ_r , safe logit ϕ_s , global budget K , max length $T_{\max}$, prompt x , parameters α, β .
 - 2: **Output:** generation $y = (y_0, \dots, y_t)$, where $t < T_{\max}$.
 - 3: **Init:** local cap $k \leftarrow K/T_{\max}$; cumulative expenditure $A_0 \leftarrow 0$; history $y_{<1} \leftarrow \emptyset$; clip $\leftarrow 0$;
 - 4: **for** $t = 0$ to $T_{\max} - 1$ **do**
 - 5: **Adaptive C:** $C \leftarrow \text{CFromKLBound}(\alpha, \beta, A_t)$.
 - 6: **Clip:** Clip the difference, $\text{clip} \leftarrow \text{clip}_C(\phi_{\text{risky}} - \phi_{\text{safe}})$.
 - 7: **Obtain $\phi_\star$:** Get $\phi_\star(\cdot \mid y_{<t}, x) \leftarrow \alpha\phi_{\text{safe}} + \beta\text{clip}$.
 - 8: **Softmax:** $p_t^* \leftarrow \text{softmax}(\phi_\star)$.
 - 9: **Sample:** $y_t \sim p_t^*(\cdot \mid y_{<t}, x)$.
 - 10: **Accrue budget:** $k_t \leftarrow \max(0, k(t+1) - A_t)$.
 - 11: **Bank realized spend:** $A_{t+1} \leftarrow A_t + D_{KL}(p_t^* \parallel p_s)$.
 - 12: **if** $y_t = \text{EOS}$ **then**
 - 13: **break**
 - 14: **end if**
 - 15: **end for**
 - 16: **Return** $y = (y_0, \dots, y_t)$.
-

Adaptive Banking. The Algorithm above uses a helper function, CFromKLBound. It implements the adaptive banking strategy, illustrated by He et al. [2], in the context of Logit Aggregate algorithm. It tracks realized spend and rolls forward unused budget from "easy" steps to overcome the overly conservative nature of a constant local cap of k . From 3.1.1, we bound the KL divergence for the per token step as follows,

$$D_{KL}(p_t^* || p_s) \leq \frac{\beta^2 C^2 \alpha}{2\tau^2}$$

From He et al. [2], we adaptively assign per-token budget in the following manner

$$k_t := \max \left(0, (t+1)k - \sum_{i=0}^{t-1} a_i \right)$$

setting the bound to the local cap k_t , we get C in terms of α, β, τ as follows

$$C = \frac{\tau}{\beta} \sqrt{\frac{2k_t}{\alpha}}$$

Note that for this choice of C , the original constraint on the global privacy budget still holds i.e, the resulting autoregressive sequence distribution $p^*(y_{<T}|x) = \prod_{t=0}^{T-1} p_t^*(y_t|y_{<t}, x)$ satisfies

$$D_{KL}(p^* || p_s) \leq K, \text{ for any } T \leq T_{\max}$$

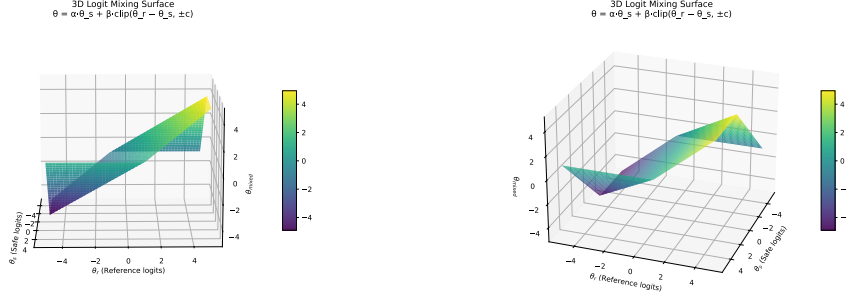


Figure 1: Visualization of Φ given Φ_s and Φ_r . Hyperparameters: $\alpha = 1$; $\beta = 1$; $c = 4$

3.1 Objective function

The expression (6) is an optimum solution of a constraint optimization problem. It can be reformulated as follows:

$$\Phi = \alpha\Phi_s + \delta; \text{ where } \delta \in [-\beta c, \beta c]; \delta = \beta(\Phi_r - \Phi_s) \text{ if } -\beta c < \delta < \beta c$$

The characteristics are as follows:

- Start at a baseline $\alpha\Phi_s$
- Add a bounded correction δ
- Update in direction towards Φ_r

A natural leads to an optimization of the form:

$$\text{define } \Phi^* = (\alpha - \beta)\Phi_s + \beta\Phi_r; \Phi = \operatorname{argmin}_{\Phi} \frac{1}{2}(\Phi - \Phi^*)^2 \text{ s.t. } |\Phi - \alpha\Phi_s| \leq \beta c$$

This can be interpreted as follows:

The optimization seeks a mixed logit Φ that minimized distance (here, euclidean) to the desired mix of safe and risky logits while constraining its deviation from the safe anchor logit $\alpha\Phi_s$ to remain within a bounded safe region of radius βc .

Or in a more geometric view: it projects a risky correction around a safe anchor. This risky projection can be tuned for high utility while maintaining proximity to the safe baseline.

3.1.1 K-NAF guarantees

Let P_* and Q be probability distribution got by the softmax over the Logits ϕ and ϕ_s . Define Q^α be the distribution for safe logit scaled by α i.e. $\text{softmax}(\alpha\phi_s)$

$$\begin{aligned} Q_i^\alpha &= \frac{Q_i^\alpha}{\sum_j Q_j^\alpha}; D_{KL}(P_*||Q) = \mathbf{E}_{P_*} \left[\log \frac{P_* Q^\alpha}{Q^\alpha Q} \right] \\ D_{KL}(P_*||Q) &= D_{KL}(P_*||Q^\alpha) + \mathbf{E}_{P_*} \left[\log \frac{Q^\alpha}{Q} \right]; \log \frac{Q^\alpha}{Q} = \log \frac{Q_i^{\alpha-1}}{\sum_j Q_j^\alpha} \\ \mathbf{E}_{P_*} \left[\log \frac{Q^\alpha}{Q} \right] &= (\alpha - 1) \mathbf{E}_{P_*} [\log Q] - \log \sum_j Q_j^\alpha \end{aligned}$$

For $\alpha \geq 1$

$$\mathbf{E}_{P_*} \left[\log \frac{Q^\alpha}{Q} \right] \leq -\log \sum_j Q_j^\alpha = (\alpha - 1) H_\alpha(Q)$$

$$D_{KL}(P_*||Q) \leq D_{KL}(P_*||Q^\alpha) + (\alpha - 1) H_\alpha(Q)$$

Here $H_\alpha(Q)$ is the Renyi entropy of Q

Now for simplicity, take Set $\alpha = 1$. Then logit difference between p_* and p_{safe} ,

$$\Delta\phi = \beta \text{clip}_C(\phi_{\text{risky}} - \phi_{\text{safe}})$$

where $\text{clip}_C(z)$ clips z such that $\|z\|_\infty \leq C$. This gives:

$$\|\Delta\phi\|_\infty \leq \beta C$$

As this difference is bounded, we obtain the following ρ -zCDP guarantees for the exponential mechanism with temperature τ :

$$\rho_{\text{token}} = \frac{1}{2} \left(\frac{\|\Delta\phi\|_\infty}{2\tau} \right)^2 \leq \frac{1}{2} \left(\frac{\beta C}{2\tau} \right)^2 = \frac{\beta^2 C^2}{8\tau^2} \quad (3)$$

Composing the mechanism over T timesteps (generating T tokens) yields the total ρ -zCDP guarantee:

$$\rho \leq T \cdot \rho_{\text{token}} = \frac{T\beta^2 C^2}{2\tau^2}$$

We convert this to a bound on the **Rényi Divergence**:

$$R_\alpha(P_*||P_{\text{safe}}) \leq \rho\alpha = \frac{T\beta^2 C^2}{2\tau^2} \cdot \alpha$$

3.1.2 Evaluation metrics

The KL Divergence between the probability distribution of the output text and the risky distribution, $D_{KL}(p_*||p_r)$ is a convenient proxy for the utility of the model. To help us keep track of this, we log both the per token KL Divergence and the global KL Divergence.

4 Experiments

4.1 Utility

We focus on two types of general utility: *fluency* and *factuality*.

Fluency aims to measure how natural and free-flowing the output reads as. We emulate the approach outlined in Chen et al. [4] and evaluate the quality of CopyBench continuations using Prometheus-v2, an LLM-as-a-judge framework that was introduced in Kim et al. [5]. This is used to score output along a 5-point rubric (5 being the highest).

Factuality is a property that we wish to preserve as factual information is not protected by copyright law. Thus, long-form factuality is evaluated in the domain of Biography, Finance, History, Science and Technology via generating outputs on select topics from these domains and scoring them after verification by a judge LLM based on known facts retrieved via web searches. [6].

4.1.1 Baselines

Single-model baselines. We use TinyComma-1.8b as our safe model that is trained on open-license data and Llama3.1-8b as our high utility risky model that is trained on copyrighted texts.

Joint-model baselines. *AnchoredDecoding* being the current state-of-the-art algorithm for copyright-free text generation was chosen as the benchmark for our algorithm. As it outperforms its predecessors on nearly all fronts, we decided to use this as the only reference baseline for our model.

4.1.2 Model pairs

We instantiate p_s and p_r as base LMs that are not fine-tuned for instructions to isolate any effects of memorization. For the risky model p_r , we chose Llama 3.1 8B and for the corresponding safe model p_s , we choose TinyComma 1.8B. We make this choice as both of them use the same Llama tokenizer.

4.1.3 Hyperparameters

We sweep all our methods across a range of k_t : $\{0.025, 0.1, 0.5, 2, 10\}$ and set the maximum length of generated text $T_{\max} = 128$ tokens.

5 Results

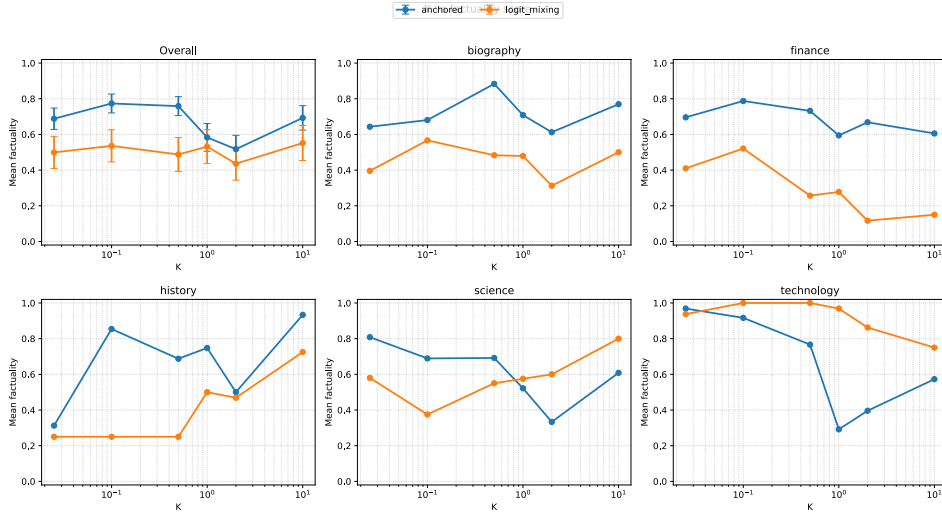


Figure 2: Comparison of Factuality Score trends

This provides the evidence that Logit Mixing and anchor decoding have different performance for the same copyright guarantees over different factuality tasks. This can be explained by the different optimization ideas these two algorithm have followed.

Overall, for Logit Mixing performs worse than anchor decoding for the same copyright guarantees in the factuality metric. This deficit in performance decreases as we loosen the copyright budget.

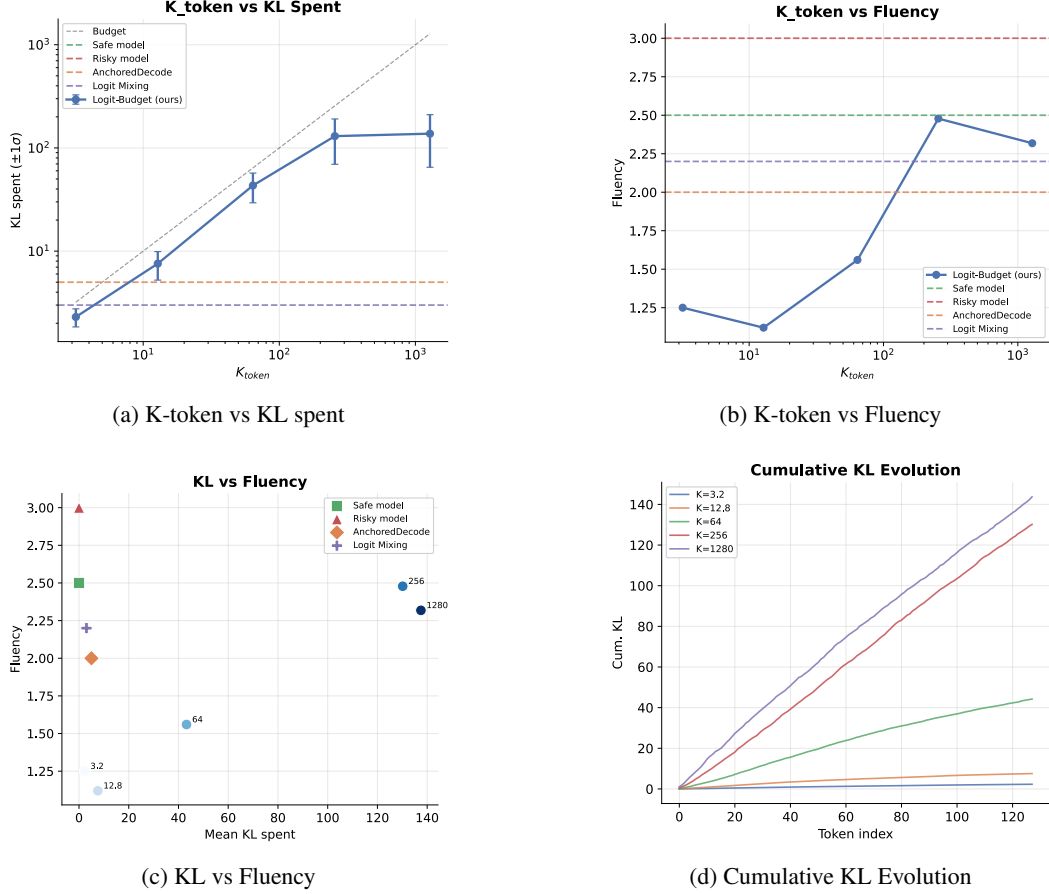


Figure 3: Comparison of fluency and KL-related metrics

Figure 3b shows that fluency improves with increasing K , with the largest gains occurring in the intermediate budget range. Beyond this region, the fluency improvements begin to saturate despite continued increases in KL expenditure.

The relationship between KL expenditure and fluency is further highlighted in Figure 3c. Higher fluency is generally associated with larger KL divergence from the safe model, suggesting an inherent tradeoff between preserving safety constraints and achieving more natural or expressive generations.

Finally, Figure 3d demonstrates the evolution of cumulative KL divergence across decoding steps for different budgets. Larger values of K accumulate divergence much more rapidly, whereas smaller budgets maintain controlled growth throughout generation.

Overall, these experiments indicate that larger decoding budgets improve fluency at the cost of increased divergence from the safe model, highlighting the controllable tradeoff between generation quality and copyright-risk mitigation.

References

- [1] Nikhil Vyas, Sham M. Kakade, and Boaz Barak. On provable copyright protection for generative models. In *International Conference on Machine Learning*, pages 35277–35299. PMLR, 2023.
- [2] Jacqueline He, Jonathan Hayase, Wen tau Yih, Sewoong Oh, Luke Zettlemoyer, and Pang Wei Koh. Anchored decoding: Provably reducing copyright risk for any language model, 2026.
- [3] Vishnu Vinod, Krishna Pillutla, and Abhradeep Guha Thakurta. Invisibleink: High-utility and low-cost text generation with differential privacy. In *NeurIPS*, 2025.
- [4] Tong Chen, Akari Asai, Niloofar Miresghallah, Sewon Min, James Grimmermann, Yejin Choi, Hannaneh Hajishirzi, Luke Zettlemoyer, and Pang Wei Koh. CopyBench: Measuring literal and non-literal reproduction of copyright-protected text in language model generation. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 15134–15158, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.844. URL <https://aclanthology.org/2024.emnlp-main.844/>.
- [5] Seungone Kim, Juyoung Suk, Shayne Longpre, Bill Yuchen Lin, Jamin Shin, Sean Welleck, Graham Neubig, Moontae Lee, Kyungjae Lee, and Minjoon Seo. Prometheus 2: An open source language model specialized in evaluating other language models, 2024. URL <https://arxiv.org/abs/2405.01535>.
- [6] Yixiao Song, Yekyung Kim, and Mohit Iyyer. Veriscore: Evaluating the factuality of verifiable claims in long-form text generation. *arXiv preprint arXiv:2406.19276*, 2024.
- [7] James Flemings, Meisam Razaviyayn, and Murali Annamaram. Differentially private next-token prediction of large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 4390–4404. Association for Computational Linguistics, 2024.
- [8] Javier Abad, Konstantin Donhauser, Francesco Pinto, and Fanny Yang. Copyright-protected language generation via adaptive model fusion. In *ICLR*, 2025.
- [9] Antonia Karamolegkou, Jiaang Li, Li Zhou, and Anders Søgaard. Copyright violations and large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7403–7412. Association for Computational Linguistics, 2023.
- [10] Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, 'Ulfar Erlingsson, Alina Oprea, and Colin Raffel. Extracting training data from large language models. In *30th USENIX Security Symposium (USENIX Security 21)*, pages 2633–2650. USENIX Association, 2021.
- [11] Mitchell Wortsman, Gabriel Ilharco, Samir Ya Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig Schmidt. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162, pages 23965–23998. PMLR, 2022.
- [12] Niva Elkin-Koren, Uri Hacohen, Roi Livni, and Shay Moran. Can copyright be reduced to privacy? In *Proceedings of the 40th International Conference on Machine Learning*. PMLR, 2023.
- [13] H. Wang et al. The stronger the diffusion model, the easier the backdoor. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235. PMLR, 2024.
- [14] Dongliang Guo, Weili Shi, and Sheng Li. LLMs are too smart to be average: Controlling LLM proficiency via guided decoding. In *ICLR 2026 (under review)*, 2025.
- [15] Parjanya Prajakta Prashant, Kaustubh Ponskshe, and Babak Salimi. Tokenswap: A lightweight method to disrupt memorized sequences in LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

- [16] Tong Chen, Faeze Brahman, Jiacheng Liu, Niloofar Mireshghallah, Weijia Shi, Pang Wei Koh, Luke Zettlemoyer, and Hannaneh Hajishirzi. ParaPO: Aligning language models to reduce verbatim reproduction of pre-training data. In *The Impact of Memorization on Trustworthy Foundation Models: ICML 2025 Workshop*, 2025. URL <https://openreview.net/forum?id=RcgdSlurDI>.
- [17] Chin-Yew Lin. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain, July 2004. Association for Computational Linguistics. URL <https://aclanthology.org/W04-1013/>.
- [18] Andrei Z. Broder. On the resemblance and containment of documents. *Proceedings. Compression and Complexity of SEQUENCES 1997 (Cat. No.97TB100171)*, pages 21–29, 1997. URL <https://api.semanticscholar.org/CorpusID:11748509>.
- [19] Sewon Min, Kalpesh Krishna, Xinxu Lyu, Mike Lewis, Wen-tau Yih, Pang Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.741. URL <https://aclanthology.org/2023.emnlp-main.741/>.